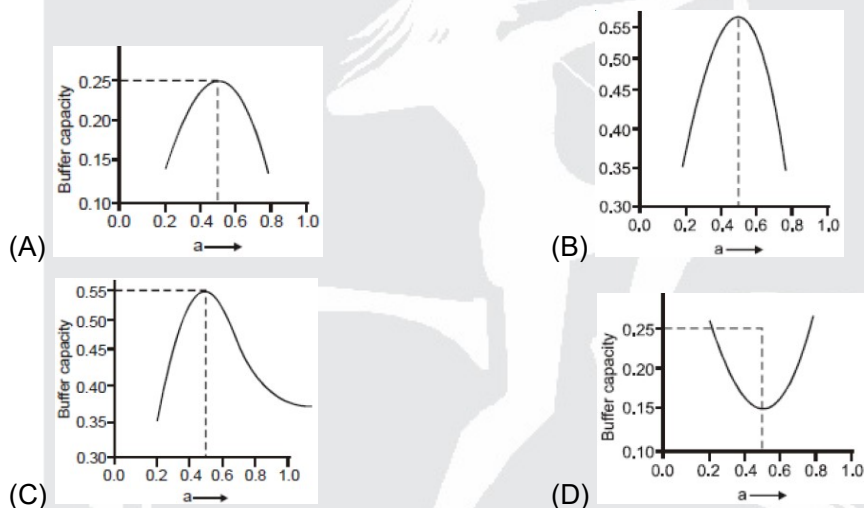


1. At what molar concentration of HCl will its aqueous solution have an $[H^+]$ to which equal contributions come from HCl and H_2O .

(A) $\sqrt{60} \times 10^{-7} M$ (B) $\sqrt{50} \times 10^{-8} M$ (C) $\sqrt{40} \times 10^{-9} M$ (D) $\sqrt{30} \times 10^{-8} M$

2. A buffer solution is prepared by mixing 'a' mole of CH_3COONa and 'b' moles of CH_3COOH into water to make 1 L buffer solution such that $(a + b) = 1$. If the instantaneous (differential) buffer capacity of this buffer solution is plotted against moles of salt CH_3COONa (a) then the plot obtained will be (to the scale) approximately. (As shown in figure in options)



3. Pure water is added into the following solutions causing a 10% increase in volume of each. The greatest % change in pH would be observed in which case (a), (b), (c) or (d)?

(A) 0.1 M $NaHCO_3$ (B) 0.2 M $NaOH$
 (C) 0.3 M $NH_3 - 0.2 M NH_4^+$ system (D) 0.4 M CH_3COONH_4

4. Liquid NH_3 dissociates to a slight extent. At a certain temp. its self dissociation constant $K_{SD}(NH_3) = 10^{-30}$. The number of NH_4^+ ions are present per $100 cm^3$ of pure liquid are:

(A) 10^{-15} (B) 6.022×10^8 (C) 6.022×10^7 (D) 6.022×10^6

5. 20 mL of 0.1 M weak acid HA ($K_a = 10^{-5}$) is mixed with solution of 10 mL of 0.3 M HCl and 10 mL of 0.1 M NaOH. Find the value of $\frac{[A^-]}{[HA] + [A^-]}$ in the resulting solution:

(A) 2×10^{-4} (B) 2×10^{-3} (C) 2×10^{-3} (D) 0.05

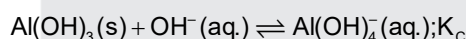
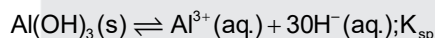
6. Which of the following expression for α of a monoacidic base (BOH) in aqueous solution at appreciable concentration is not correct?

(A) $100 \times \sqrt{\frac{K_b}{c}}$ (B) $10^{p_{OH} - p_{Kb}}$ (C) $\frac{K_w [H^+]}{K_b + K_w}$ (D) $\frac{K_b}{K_b + [OH^-]}$

7. The simultaneous solubility of AgCN ($K_{sp} = 2.5 \times 10^{-16}$) and AgCl ($K_{sp} = 1.6 \times 10^{-10}$) in 1.0 M NH_3 (aq) are respectively: [Given: $K_f [Ag(NH_3)_2^+] = 10^7$]

(A) $0.037, 5.78 \times 10^{-8}$ (B) $5.78 \times 10^{-8}, 0.037$ (C) $0.04, 6.25 \times 10^{-8}$ (D) $1.58, 1.26 \times 10^{-5}$

8. The $Al(OH)_3$ is involved in the following two equilibria,



Which of the following relationship is correct at which solubility is minimum?

(A) $[OH^-] = \left(\frac{K_{sp}}{K_c} \right)^{1/3}$ (B) $[OH^-] = \left(\frac{K_c}{K_{sp}} \right)^{1/4}$
 (C) $[OH^-] = \sqrt{\left(\frac{K_{sp}}{K_c} \right)^{1/4}}$ (D) None of these

MULTIPLE CHOICE QUESTIONS

9. If K_{a_1}, K_{a_2} and K_{a_3} be the first, second and third ionization constant of H_3PO_4 and $K_{a_1} \gg K_{a_2} \gg K_{a_3}$ which is/are correct :

(A) $[H^+] = \sqrt{K_{a_1} [H_3PO_4]}$ (B) $[H^+] = [HPO_4^{2-}]$
 (C) $K_{a_2} = [HPO_4^{2-}]$ (D) $[HPO_4^{2-}] = [PO_4^{3-}]$

10. Which of the following mixtures constitute a buffer ?



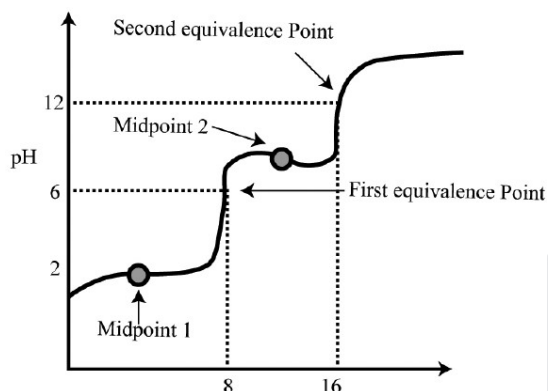
11. Which of the following statements is/are correct?

- (A) The conjugate acid of NH_2^- is NH_3
 (B) Solubility product constant increases with increase in concentration of ions
 (C) On diluting a buffer solution pH change is negligible
 (D) In alkaline buffer solution, If some HCl is added, its $[OH^-]$ will increase

12. Formic acid is a weak acid and hydrochloric acid is a strong acid. It follows that the :

- (A) $[OH^-]$ of 0.01 M HCl (aq.) will be less than that of 0.01 M HCOOH (aq.)
 (B) solution containing 0.1 M NaOH(aq.) and 0.1 M HCOONa(aq.) is a buffer solution
 (C) pH of 10^{-9} M HCl (aq.) will be approximately 7 at $25^\circ C$
 (D) pH of a solution formed by mixing equimolar quantities of HCOOH and HCl will be less than that of a similar solution formed from HCOOH and HCOONa

13. A weak acid (or base) is titrated against a strong base (or acid), volume v of strong base (or acid) is plotted against pH of the solution (as shown in figure). The weak electrolyte (i.e., acid or base) could be



- (A) Na_2CO_3 (B) $\text{Na}_2\text{C}_2\text{O}_4$ (C) $\text{H}_2\text{C}_2\text{O}_4$ (D) $\text{CH}_2(\text{COOH})_2$
14. Which of the following statement(s) is/are correct about the ionic product of water
 (A) K_i (ionization constant of water) $< K_w$ (ionic product of water)
 (B) $\text{p}K_i > \text{p}K_w$
 (C) At 25°C , $K_i = 1.8 \times 10^{-14}$
 (D) Ionic product of water at 10°C is 10^{-14}
15. H_2A is a weak diprotic acid. If the pH of 0.1M H_2A solution is 3 and concentration of A^{2-} is 10^{-12} at 25°C .
 Select correct statement(s)?
 (A) $[\text{H}^+]_{\text{total}} \approx [\text{H}^+]$ from first step of ionization of acid H_2A
 (B) Concentration of OH^- in solution is 10^{-3}M
 (C) The value of K_{a_1} is nearly 10^{-5}
 (D) $\text{p}K_{a_2} = \text{p}K_{a_1} = 9$
16. 0.01M NH_4Cl (aq.) solution at 25°C has:
 (A) $[\text{Cl}^- \text{ (aq.)}] < 10^{-2}\text{M}$ (B) $[\text{NH}_4^+ \text{ (aq.)}] < 10^{-2}\text{M}$
 (C) $\text{pOH} > 7$ (D) $[\text{H}^+] > 10^{-7}\text{M}$

COMPREHENSION # 1 (FOR Q. 17)

17. Potash alum is $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$. As a strong electrolyte, it is 100 % dissociated into K^+ , Al^{3+} and SO_4^{2-} . The solution is acidic because of the hydrolysis of Al^{3+} , but not so acidic as might be expected, because the SO_4^{2-} can sponge up some of the H_3O^+ by forming HSO_4^- .
 Given a solution made by dissolving 11.85 gm of $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$ in enough water to make 1000 cm^3 of solution. What is $[\text{H}_3\text{O}^+]$ of the solution if Both Al^{3+} and SO_4^{2-} are hydrolysing.
 First hydrolysis constant for Al^{3+} is 1.4×10^{-5}
 Acid dissociation constant for HSO_4^- in water is 1.25×10^{-2}
 (A) $2.93 \times 10^{-4}\text{M}$ (B) 0.0114M (C) $5.43 \times 10^{-6}\text{M}$ (D) None of these

18. Match the column

Column I (pH of resultant solution)

- (A) 200 ml of H_2SO_4 solution (specific gravity 1.225 containing 25% H_2SO_4 by weight) + 800 ml of 0.525 M strong triacidic base $\text{X}(\text{OH})_3$
- (B) 50 ml of 0.1 M HCO_3^- + 50 ml of 0.8 M CO_3^{2-}
($\text{H}_2\text{CO}_3; K_{a_1} = 4 \times 10^{-7}, K_{a_2} = 2 \times 10^{-11}$)

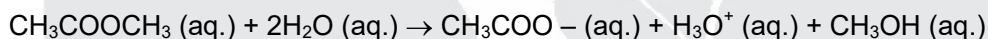
- (C) 50 ml of 0.2 M HA (aq) ($K_a = 10^{-5}$) + 50 ml of 0.1 M HCl (aq) + 100 ml of 0.13 M NaOH (aq)

Column II (Exist between Colour transition range of an indicator)

- (P) Phenol Red (6.8 to 8.4)
- (Q) Propyl red (4.6 to 6.4)
- (R) Phenolphthalein (8.3 to 10.1)
- (S) Malachite green (11.4 to 13)

SUBJECTIVE ANSWER TYPE

19. An $\text{NH}_4^+ - \text{NH}_3$ buffer is supposed to keep the pH of the solution constant within 0.3 pH unit during the reaction.



If this solution had initial concentrations: $[\text{NH}_4^+]_0 = 0.1 \text{ M}$, $[\text{NH}_3]_0 = 0.06 \text{ M}$, $[\text{CH}_3\text{COOCH}_3]_0 = 0.02 \text{ M}$.

What would be the initial and final pH of the solution? Is this a satisfactory buffer? [$K_b(\text{NH}_3) = 1.8 \times 10^{-5}$, $\log 2 = 0.3$, $\log 3 = 0.48$].

20. A solution of 0.1 M Cl^- , 0.1 M Br^- and 0.1 M I^- solid AgNO_3 is gradually added to this solution. If the addition of AgNO_3 does not change the volume. Answer the following:

- (i) What conc. of Ag^+ ions will be required to start precipitation of each of the three ions.
- (ii) Which ion will precipitate first
- (iii) What will be the conc. of this ion when the second ion start precipitating.
- (iv) What will be the conc. of both ions when the third ion start precipitating.

Given: $K_{sp}(\text{AgCl}) = 1.7 \times 10^{-10}$, $K_{sp}(\text{AgBr}) = 5 \times 10^{-13}$, $K_{sp}(\text{AgI}) = 8.5 \times 10^{-17}$.

21. Calculate the solubility of AgCN in a buffer solution of pH = 3. Neglect any complexation. Take $K_{sp}(\text{AgCN}) = 3.2 \times 10^{-16}$, $K_a(\text{HCN}) = 6.4 \times 10^{-10}$.

22. If 0.00050 mol NaHCO_3 is added to 1 litre of a buffered solution at pH 8.00, how much material will exist in each of the three forms H_2CO_3 , HCO_3^- and CO_3^{2-} ? For H_2CO_3 , $K_1 = 5 \times 10^{-7}$, $K_2 = 5 \times 10^{-13}$.

23. Equilibrium constant for the acid ionization of Fe^{3+} to $\text{Fe}(\text{OH})^{+2}$ and H^+ is 6.5×10^{-3} . What is the max. pH, which could be used so that at least 95% of the total Fe^{3+} in a dilute solution exists as Fe^{3+} ?

24. Predict whether or not AgCl will be precipitated from a solution which is 0.02 M in NaCl and 0.05 M in $\text{KAg}(\text{CN})_2$. Given $K_{\text{diss}}(\text{Ag}(\text{CN})_2^-) = 4.0 \times 10^{-19} \text{ M}^2$ and $K_{sp}(\text{AgCl}) = 2.8 \times 10^{-10} \text{ M}^2$.